

Managing personal finances is a big challenge for young people, especially college students and people in their 20s who are trying to become financially independent. Many young people have irregular incomes, rely on part time work or financial aid, and face regular expenses such as rent, food, subscriptions, and transportation. While most banks offer transaction history and basic budgeting tools, these systems often display information without interpretation, guidance, or customization. As a result, users often know they're wasting money, but they don't understand why it's happening or how they can realistically change their behavior. The real business issue addressed in this project is the lack of smart, adaptive financial advice that helps young people understand their spending habits, plan sustainable budgets, and make informed financial decisions.

The target audience for this system is young people aged 18 to 25, mainly students. These users typically have limited financial experience and are unwilling or unable to work with professional financial advisors. They want tools that are accessible, nonjudgmental, explain financial information in simple terms and tailored to their personal goals and limitations. An agent based AI solution is well suited to solve this problem because personal finance is not a single step task, but an ongoing process that involves cleaning, analyzing, planning, explaining and reviewing data over time. Unlike static budgeting applications or fixed human designed workflows, agent based systems can dynamically coordinate multiple specialized agents to interpret data, generate insights, and adjust recommendations as user behavior changes. A common use case for this system occurs when a user wants to know why they don't have enough money by the end of the month. In this scenario, the user downloads a bank transaction file or manually enters their recent expenses and monthly income. The system first cleans and organizes transaction data, then categorizes expenses into meaningful groups such as food, rent, entertainment and subscriptions.

Analytical agents identify patterns and highlight problem areas such as unusually high food delivery costs or recurring charges that users may have overlooked. Your coach will then explain these results in clear terms and suggest realistic adjustments, such as setting weekly limits or cutting back on certain discretionary expenses.

The second scenario assumes that the user wants to create a realistic monthly budget and savings plan. You enter information about your income and financial goals, such as saving a certain amount or reducing your credit card usage, and the system generates a personalized budget based on your past spending behavior rather than ideal assumptions. The planning officer allocates funds into categories, taking into account fixed and variable costs. The system can simulate alternative scenarios to show how small changes in behavior affect long term results. Reviewers consider budget feasibility and ensure recommendations are appropriate for youth with limited income. The third scenario focuses on continuous financial monitoring and adjustment. As the user continues to add new transactions or update financial goals, the system evaluates whether it is within the budget. If a deviation occurs, the agent's workflow reevaluates priorities and provides updated recommendations. Instead of issuing general warnings, the system explains the consequences of recent spending decisions and provides specific recommendations tailored to the user's lifestyle. This continuous feedback loop helps users develop financial awareness and healthier habits over time. The initial system design consists of multiple specialized agents working together under the direction of a coordinator agent. The data cleansing agent is responsible for preprocessing transaction data, handling missing values, and normalizing merchant names. Spend analytics agents categorize spending and identify trends and anomalies. Budget planners create personalized financial plans and evaluate trade offs between spending and saving. Behavioral coaching agents communicate ideas and recommendations in a

supportive, nonjudgmental tone. A reviewer or safety officer reviews all results to ensure that recommendations are reasonable, prudent, and appropriate for young users. The coordinator agent manages the workflow and ensures consistency of agent output.

The system uses local data sources such as transaction history and uploaded CSV files containing manually entered income and financial goals. A simple front-end interface allows users to upload data, enter settings, and view a visual overview of their spending and budget. The visualization tool shows spending breakdowns and budget comparisons, but direct banking APIs are avoided to protect user privacy. All data processing occurs locally or in the application environment.

Feasibility studies are expected to encounter a number of issues, including the risk of inconsistent or incomplete financial data and inaccurate or overly aggressive financial recommendations. These issues are addressed through critical agents that provide data validation, conservative assumptions, and realistic and safe recommendations. Extensive transaction history is managed using aggregation technology and no sensitive personal data is stored after each session. The project will be carried out according to the course schedule from system design to front end development, multi-agent workflow implementation, and final evaluation, and the final deliverables will be presented individually along with a working prototype and evaluation results.